

Interacting with Neural Radiance Fields in Immersive Virtual Reality

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ABSTRACT

Recent advancements in the neural radiance field (NeRF) technology, in particular its extension by instant neural graphics primitives, provide tremendous opportunities for the use of real-time immersive virtual reality (VR) applications. Moreover, the recent release of an immersive neural graphics primitives framework (immersive-ngp) brings real-time, stereoscopic NeRF rendering to the Unity game engine. However, the system and application research combining NeRF and human-computer interaction in VR is still at the very beginning.

In this demo, we present multiple interactive system features for immersive-ngp with design principles focusing on improving the usability and interactivity of the framework for small to medium-scale NeRF scenes. We demonstrate that these new feature implementations such as exocentric manipulation, VR tunneling effects, and immersive scene appearance editing enable novel VR-NeRF experiences, for example, for customized experiences in inspecting a particle accelerator environment.

CCS CONCEPTS

• **Virtual Reality**; • **Computer Graphics**; • **Neural Radiance Field**;

KEYWORDS

Immersive Virtual Reality, Neural Radiance Field

ACM Reference Format:

Ke Li, Tim Rolff, Susanne Schmidt, Reinhard Bacher, Wim Leemans, and Frank Steinicke. 2023. Interacting with Neural Radiance Fields in Immersive Virtual Reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3544549.3583920>

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583920>

1 INTRODUCTION

The rapid development of virtual reality (VR) technology enables a wide range of immersive virtual experiences through affordable commercial head-mounted displays. Despite the enormous potential of VR systems in providing realistic virtual experiences, many existing VR scene components are still based on low-fidelity polygon meshes with low resemblance to actual real-world objects. A lack of realism of VR environments cannot only decrease the user's sense of presence [15], but also limits the applicability and reliability of such VR systems, for example, for secure and high-precision robot teleoperation [7], efficient VR training and education [9], and immersive journalism [3].

Recently, Neural radiance fields (NeRF) [10, 11] have rapidly emerged as a popular 3D reconstruction method that can produce photorealistic free-view-point videos with accurate scene details [19]. The creation of a 3D NeRF scene only requires multiple 2D images and their camera poses as input to an artificial neural network. Further, 3D reconstruction through NeRF techniques demonstrate appealing advantages over many conventional 3D reconstruction methods in terms of scene realism, reconstruction quality, and hardware costs [5, 14]. Although the original NeRF method proposed by Mildenhall et al. [10] requires extensive computation for training a single scene, the development of instant neural graphics primitives (instant-ngp) with a multi-resolution hash encoding proposed by Müller et al. [11] enables rapid training and real-time NeRF rendering.

Prior to instant-ngp, most of the open-source implementations of NeRF techniques had been developed with machine learning frameworks based on high-level programming languages, such as Python [2, 10], which often provide limited rendering performance and cannot be efficiently applied to low-level real-time rendering systems such as the Unity game engine. This interoperability challenge between high-level languages and low-level graphics systems has made real-time interactive NeRF systems for immersive VR applications largely unavailable. Recently, the open-source VR-NeRF framework *immersive-ngp* was released by Li et al. [8], integrating the original low-level implementation of instant-ngp into the Unity game engine through a native dynamic link library plugin.

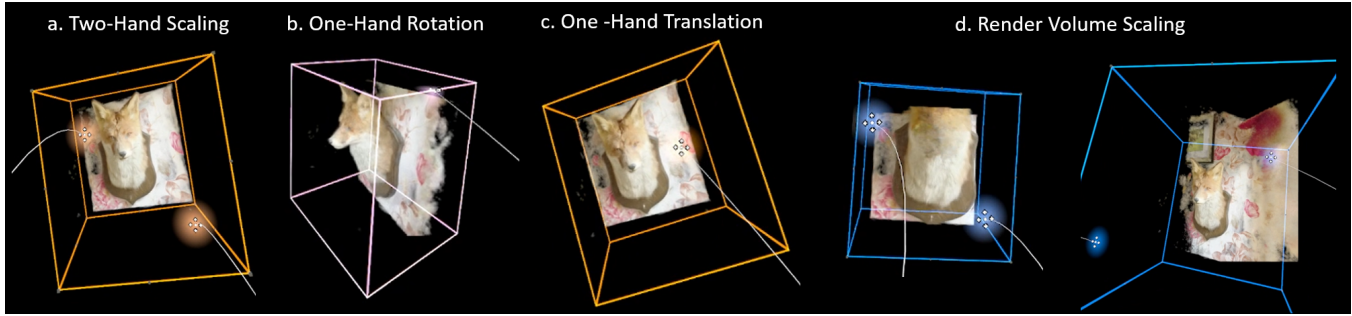


Figure 1: The design of exocentric manipulation techniques for small to medium size NeRF scenes in Unity with the Mixed Reality Toolkit (MRTK), with (a) demonstrates scene scaling involving both of the user’s hands, (b) illustrates local rotation, (c) demonstrates one hand translation manipulation, and (d) illustrates interactive render volume adjustment through scaling the bounding box.

Thus, *immersive-ngp* enables rapid prototyping and benchmarking of the *instant-ngp* technique for the human-computer interaction (HCI), computer vision (CV), and VR community. Li et al. demonstrated that small to medium-size NeRFs can already be used for high-resolution and low-latency stereoscopic rendering in real-time immersive VR experiences [8].

In this demo, we report on the current development of *immersive-ngp* and present new features with design principles focusing on improving the interactivity and usability of our system with small to medium-size NeRF scenes through techniques such as exocentric manipulation, VR tunneling, and customized scene appearance editing. For the CHI Interactivity live demo session, we prepare two installations to demonstrate these features to the broader audience. At the first workstation, participants can experiment with the current features with the standard evaluation NeRF fox dataset. Alternatively, participants could also bring their own pre-processed *instant-ngp* dataset (with small to medium-scale scenes) for live interaction and customization of their own NeRF scene, for example, to experience scene appearance editing effects. In the second installation, we present a practical use case of our system at the European X-Ray Free Electron Laser (EuXFEL) linear particle accelerator facility [4], a hazardous and radioactive environment that is inaccessible and hostile to humans during runtime. This shows the potential of *immersive-ngp*, demonstrating how an interactive immersive VR-NeRF system could be used to assist accelerator operators with remote facility inspection and maintenance.

2 FEATURE OVERVIEW

Our extension of *immersive-ngp* specifically aims to evolve the framework from pure scene visualization to interactive scene manipulation. This is essential for VR-NeRF to unfold its potential in practical use cases.

As a specific example use case, we demonstrate the interaction with a NeRF scene depicting a 10-meter section of the EuXFEL particle accelerator tunnel as illustrated in Figure 2. Traditionally, the operation of such a complex facility involves extensive onsite inspection and maintenance, which requires a scheduled shutdown of the machine to prevent radiation hazards before humans could

enter the accelerator tunnel. Not only do such onsite inspection and maintenance routines reduce the availability of the facility for conducting scientific experiments, but they also decrease the energy efficiency of accelerator operation and increase the required human resources [18]. An interactive NeRF rendering of the particle accelerator has the potential to improve inspection and maintenance efficiency by providing a realistic representation of the remote and hazardous accelerator environment.

As it is currently technically challenging to create an accurate and detailed NeRF representation of the entire 2 kilometers accelerator tunnel, the facility could be divided into individual small sections. For such scenes, the user can utilize the following proposed features, such as exocentric manipulation for rapid perspective change, motion sickness reduction through VR tunneling, and render volume adjustment to focus on specific scene content.

2.1 Interactive Exocentric Manipulation

The initial release of *immersive-ngp* allows six degrees of freedom (6-DoF) egocentric navigation within a NeRF scene. This enables the user to freely move through the scene based on input from the VR controllers’ joysticks and head movements and provides an intuitive way for traveling within a scene. However, there are other possibilities for the inspection of a NeRF scene from different viewpoints or the alignment with generated meshes or backgrounds without the need to navigate through the scene. Here, exocentric manipulation techniques can be beneficial for the natural transformation of a NeRF scene. As Figure 1 illustrates, we implemented interaction techniques for four different types of exocentric interaction: scaling transformation, local rotation transformation, translation transformation, and render volume scaling. As NeRF is based on volumetric rendering, transforming the NeRF scene in object space requires the pivot point to be set to the center of the render volume, which is defined as the axis-aligned bounding box (AABB) [1] in the original *instant-ngp* implementation. Similarly, we define a virtual bounding box in Unity and utilize the popular mixed reality user interface (UI) framework provided by Microsoft Mixed Reality Toolkit (MRTK)¹ for natural and intuitive object manipulation. In addition, this feature enables dynamic scaling of the

¹<https://github.com/microsoft/MixedRealityToolkit-Unity>

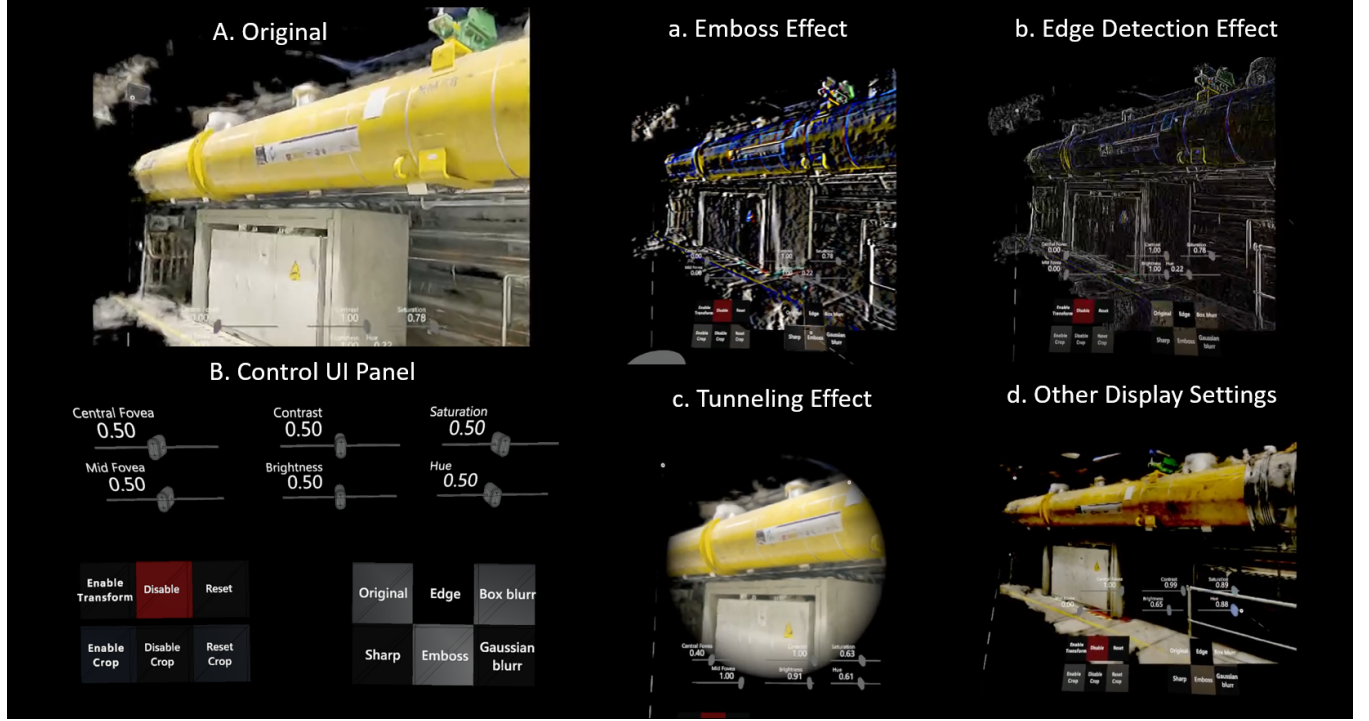


Figure 2: Screenshots of the UIs and example image filtering effects of the NeRF scene editing features applied to a scene containing 10 meters section of the European X-Ray Free Electron Laser (EuXFEL) linear particle accelerator tunnel.

render volume, allowing for cutout or magnification of the render volume depending on the performance capabilities of different GPU devices.

2.2 VR Tunneling

The initial benchmarking results of immersive-ngp [8] reveal that a medium-size render volume filled with complex scene details can lead to an increase in GPU and CPU frame times and a decrease in frame rate due to the increase in the number of network queries performed for volume rendering. This could lead to the undesired effect of VR motion sickness due to the larger motion-to-photon latency. Therefore, we integrate a VR tunneling effect, a popular method in active VR games to reduce the user’s motion sickness by applying a field of view (FoV) restriction at the peripheral region of the human eyes [6]. Figure 2 (c) illustrates the tunneling effect applied to the standard evaluation NeRF fox scene in immersive VR, where the user can freely adjust the display FoV of the NeRF scene.

2.3 Immersive NeRF Appearance Editing

The volumetric rendering nature of NeRF exhibits a major challenge in changing the the appearance of the original NeRF scene in the traditional graphics rendering pipeline due to the missing geometric mesh details. Appearance editing, however, would be an important feature to allow users artistic freedom in designing and customizing their NeRF scenes. Although advanced appearance editing techniques, such as style-transfer [12], require training and

inference on a separate neural network, customized visual effects for NeRF appearance editing can also be achieved with image filter effects based on Unity shaders. For this purpose, we adapt the image effects and filter package provided by Ashwin² and enable interactive NeRF appearance editing in VR.

Figure 2 presents the UIs and example effects of the appearance editing application. The UIs include several buttons which allow users to directly apply specific image filters to the render plane of the NeRF images, such as edge, box blur, sharpen, emboss, and gaussian blur effects. The sliders allow users to customize the Hue Saturation Value (HSV) and contrast value with VR controllers or through hand tracking. The control panels are positioned at the lower vertical peripheral region to avoid distracting the user’s workflow.

3 SYSTEM SETUP

Similar to the first release of immersive-ngp [8], our system only requires a moderate gaming-grade laptop, with an RTX 2080, for smooth 20–30fps when displaying medium-scale NeRF scenes (50–100 images). As our framework is based on SteamVR working together with the legacy OpenVR plugin in the Unity game engine, it is compatible with many VR headsets, including Meta Quest 2, Meta Quest Pro, Vive Pro, and Vive Pro Eye. At the live demo session, we present the framework at maximum performance by using a workstation equipped with an Nvidia RTX 3090 GPU and a Meta Quest Pro VR headset for the particle accelerator scene with 251

²<https://assetstore.unity.com/packages/vfx/shaders/image-effects-filters-92430>

images as input and 10 meters long render volume. For the second custom evaluation installation, we prepare an affordable setup with an Alienware laptop with an RTX 2080 GPU and a Meta Quest 2 VR headset. This allows participants to evaluate different VR-NeRF experiences based on both high-performance and moderate graphics devices. Both immersive experiences can be provided in a small stationary seated or standing area.

4 CONCLUSION

In this demo, we extended the original immersive-ngp framework and present new system features for improving the system usability and interactivity when manipulating NeRF scenes in VR. The new features enable 6-DoF exocentric manipulation, VR tunneling effects, and immersive appearance editing of any small to medium-scale instant-ngp scene, and could run on a gaming-grade laptop and a commercial VR headset. We demonstrate the applicability of our system with example scenes from a particle accelerator tunnel, a radioactive and hazardous environment where immersive NeRF technology could improve the operation efficiency and operation safety of the facility. As the described features will be integrated into immersive-ngp as future open-source releases, our system will also open up new possibilities for novel photorealistic VR experiences in other application domains, such as immersive VR shopping [13] and interactive virtual exhibitions [17], or to provide a possible extension for gaze mapping [16].

ACKNOWLEDGMENTS

This work was supported by DASHH (HELMHOLTZ Graduate School for the Structure of Matter) with the Grant-No. HIDSS-0002, the German Federal Ministry of Education and Research (BMBF), and the German Research Foundation (DFG).

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